

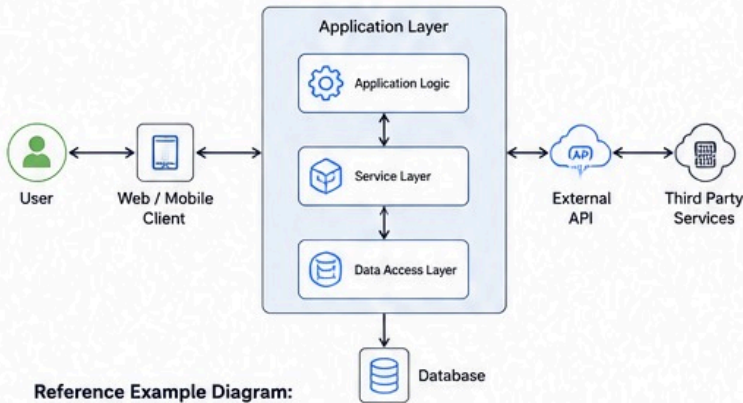
Project Design Phase-II Technology Stack (Architecture & Stack)

Date	27 June 2026
Team ID	RACHANA-TEAM
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2.

Example: Order processing during pandemics for offline mode



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Reference Example Diagram:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular.js / React.js etc.

2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	ServiceNow Flow / Script Include
4.	Application Logic-3	Logic for a process in the application	Business Rule / Workflow
5.	Database	Data Type, Configurations etc.	MySQL / PostgreSQL
6.	Cloud Database	Database Service on Cloud	AWS RDS / Azure SQL
7.	File Storage	File storage requirements	AWS S3 / Azure Blob Storage
8.	External API-1	Purpose of External API used in the application	REST API
9.	External API-2	Purpose of External API used in the application	Third Party API
10.	Machine Learning Model	Purpose of Machine Learning Model	NLP / Classification Model
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration, Cloud Server Configuration	AWS / Azure / On-Prem Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Spring Boot, Node.js, React, etc.
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	OAuth 2.0, JWT, RBAC, HTTPS, etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Microservices / 3-Tier Architecture
4.	Availability	Justify the availability of application (e.g., use of load balancers, distributed servers etc.)	Load Balancer, Auto Scaling, etc.

S.No	Characteristics	Description	Technology
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's etc.)	Redis Cache, CDN, AWS CloudFront

References (for Quality & Architecture Best Practices):

- <https://www.ibm.com/cloud/architecture>
- <https://docs.servicenow.com/>
- <https://aws.amazon.com/architecture/>
- <https://learn.microsoft.com/en-us/azure/architecture/>
- <https://microservices.io/patterns/>